

A Deep Learning Model for the diagnosis of Gastric Neuroendocrine Carcinoma

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

General Comments

This paper attempts to classify neuroendocrine carcinoma vs conventional gastric carcinoma. The research has high utility from the perspective of correctly suggesting diagnoses that pathologists might overlook.

Major Comments

1. The model reported in this paper performs "classification" between NEC and conventional GC rather than "detection" of NEC. This distinction is potentially misleading in several sections, including the abstract, which suggest detection capabilities. To properly claim NEC detection, validation would need to be performed using the entire population of gastric cases as the denominator. This represents a significant issue affecting clinical utility and should be addressed in the limitation. Additionally, it would be beneficial to suggest specific use cases, such as combining this model with gastric cancer detection models or implementing it as a diagnostic assistance tool.
2. There is confusion in the description of internal and external cohorts:
 - The text mentions "external cohort sourced from multiple medical centers" (line 112), but Figure S1 and Table S1 indicate that the cohort is from Sun Yat-sen University. Please clarify which is correct.
 - If both the internal and external cohorts are from Sun Yat-sen University, describing one as "external" is misleading since they share the same data source.
 - The handling of the internal cohort appears in both cross-validation and test subset contexts, which is confusing for readers.
 - For clarity, the reviewer may suggest renaming the current "internal (test subset)" as "internal test cohort 1" and the current "external cohort" as "internal test cohort 2." (if my understanding is correct)
 - If cases from the same institution are used across different cohorts, please explicitly state that no patient overlap in multiple cohorts.
3. Given the rarity of gastric NET, further description is needed regarding how cases in the external cohort and consultation cohort were selected. If neuroendocrine tumors, other rare histological types, or non-neoplastic diseases were excluded, this should be clearly stated.
4. The GitHub link provided in line 292 is not functioning, making data availability currently unknown.

Minor Comments

1. Please provide the definition of NEC diagnosis used in this study (i.e. WHO 5e).
2. Please clarify whether the pathologists' diagnoses in the reader study were made using only H&E stains.
3. Lines 162-163 mention the number of tiles used. Does this refer to the number of whole slide images used? Also, please verify that the numbers allocated for training and validation are not reversed.

Reviewer #2

(Remarks to the Author)

This manuscript aims to develop a deep learning-based diagnostic model that serves as a rapid and cost-effective screening solution to aid in the diagnosis of gastric neuroendocrine carcinoma (G-NEC). The study was well-designed and logically structured, and the text was concisely written.

Strengths:

1. Three cohorts were included in the study: internal, external, and consultation-cohorts. The slides and images were collected from multiple medical centers.
2. The model's performance was compared with that of pathologists with different levels of expertise/experience, and it shows that the model's diagnosis surpasses the pathologists.
3. The inclusion of workflow figures, detailed statistics of the study participants, performance during five-fold validation, and histology images is very friendly to readers.
4. The study results support the hypothesis that the deep learning-based diagnostic model overcomes the routine diagnosis of G-NECs with the assistance of IHC and molecular tests.

Improvements:

1. The main diagnostic challenge is to differentiate G-NEC from poorly differentiated gastric adenocarcinoma. The differentiation grades were listed in Supplementary Table 1, including poorly, moderately, and well-differentiated gastric adenocarcinoma. For the unsuccessful cases predicted by the G-NECNet, did the author review the histology to see if the cases had more ambiguities/overlapping features with gastric adenocarcinoma?
2. Did the study include any cases with combined components (both neuroendocrine carcinoma and adenocarcinoma), and how about the performance for such cases?
3. Please check the figure 3 legend, and panel d was not included.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

I appreciate the authors' efforts to properly address reviewer concerns and make the content clearer.

Regarding lines 186-188, which detail the data distribution for your 5-fold cross-validation, I noticed a potential discrepancy. In a standard 5-fold cross-validation (where four folds are used for training and one for internal validation), the training set is typically about four times larger than the validation set. However, the current description (16 G-NEC + 85 GC for training vs. 61 G-NEC + 340 GC for internal validation) seems to indicate the opposite. Could you please clarify if the numbers for the training and internal validation sets might have been inadvertently swapped, or if there's a specific reason for this distribution?

Reviewer #2

(Remarks to the Author)

I appreciate that the authors carefully addressed all the comments and suggestions from reviewers and support the publication of the revised manuscript.

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A point-by-point response to the reviewers

Referee expertise:

Referee #1: histopathology, machine learning

Referee #2: histopathology, gastrointestinal tract cancers

Reviewers' comments:

Reviewer #1 (Remarks to the Author):

General Comments

This paper attempts to classify neuroendocrine carcinoma vs conventional gastric carcinoma. The research has high utility from the perspective of correctly suggesting diagnoses that pathologists might overlook.

Our reply:

We were pleased that the reviewer felt that “The research has high utility”.

Some specific points:

Major Comments

1. The reported in this paper performs “classification” between NEC and conventional GC rather than “detection” of NEC. This distinction is potentially misleading in several sections, including the abstract, which suggest detection capabilities. To properly claim NEC detection, validation would need to be performed using the entire population of gastric cases as the denominator. This represents a significant issue affecting clinical utility and should be addressed in the limitation. Additionally, it would be beneficial to suggest specific use cases, such as combining this with gastric detection models or implementing it as a diagnostic assistance tool.

Our reply:

We sincerely thank you for this important and constructive comment. We agree that our current study focuses on classification between gastric neuroendocrine carcinoma (G-NEC) and gastric adenocarcinoma (GC), rather than on detection of G-NEC from the entire population of gastric lesions. The rationale for this design is that the most challenging and clinically relevant differential diagnosis arises between G-NEC and poorly differentiated GC, which often show overlapping histomorphology features and immunohistochemical profiles. Therefore, we intentionally restricted the dataset to pathologically confirmed GC and G-NEC cases to ensure high-quality, well-annotated training and evaluation.

We fully acknowledge that this design limits the model’s ability to perform “G-NEC

detection” in a true screening context across all gastric pathologies. As suggested, we have revised the manuscript to clarify this distinction and added a paragraph in the Discussion section to explicitly state this limitation and to propose potential future directions. Corresponding changes have been made in the Title, Abstract and Discussion to replace the term “detection” with “classification” or “diagnosis” where appropriate and to add this limitation discussion (please refer to Lines 2, 58 and 302-308 in the revised manuscript).

Within the Discussion section:

“First, although this was a multicenter study, the overall sample size remained relatively limited, and the dataset included only histologically confirmed G-NEC and gastric GC cases. This design was chosen because differentiating G-NEC from poorly differentiated GC represents the most challenging and clinically relevant diagnostic scenario in gastric pathology. However, this binary classification setting does not encompass the full spectrum of gastric lesions, such as MiNEN, other rare histological types, and non-neoplastic conditions. Expanding future studies to include a broader population would enable true “detection” of G-NEC and enhance model generalizability.”

2. There is confusion in the description of internal and external cohorts:

- The text mentions “external cohort sourced from multiple medical centers” (line 112), but Figure S1 and Table S1 indicate that the cohort is from Sun Yat-sen University. Please clarify which is correct.
- If both the internal and external cohorts are from Sun Yat-sen University, describing one as “external” is misleading since they share the same data source
- The handling of the internal cohort appears in both cross-validation and test subset contexts, which is confusing for readers.
- For clarity, the reviewer may suggest renaming the current “internal (test subset)” as “internal test cohort 1” and the current “external cohort” as “internal test cohort 2.” (if my understanding is correct)
- If cases from the same institution are used across different cohorts, please explicitly state that no patient overlap in multiple cohorts.

Our reply:

We sincerely thank you for this insightful comment. We have carefully clarified the cohort definitions and revised the corresponding descriptions in the manuscript to eliminate potential confusion. The detailed clarifications are as follows:

1) Data sources:

The internal cohort was collected from the Sun Yat-sen University Cancer Center (SYSUCC). The so-called external cohort was collected from the Sixth Affiliated Hospital of Sun Yat-sen University (SYSU-6th Hospital). Although both hospitals are affiliated with Sun Yat-sen University, they are independent medical institutions with distinct patient populations, diagnostic systems, and pathology archives. Therefore, there was no overlap or data sharing between their patient cohorts.

2) Internal cohort handling:

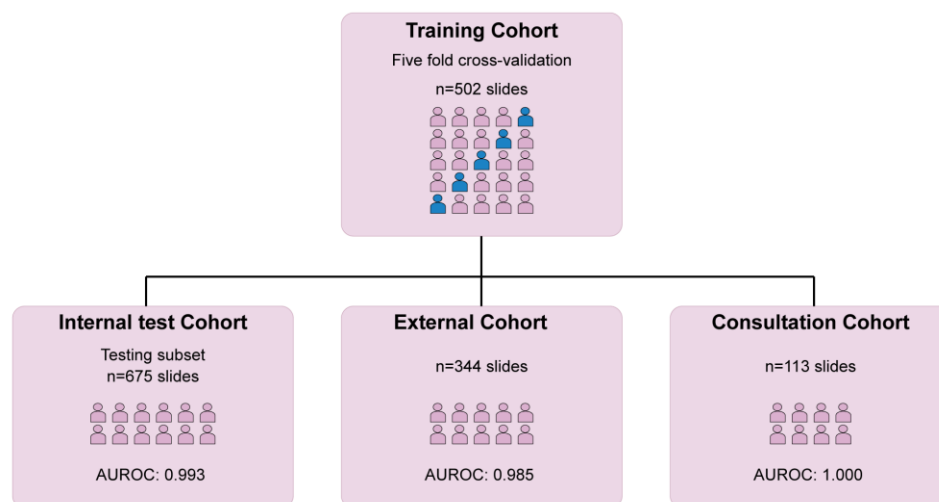
Within the internal cohort (SYSUCC), cases were randomly divided into two mutually exclusive subsets — a training set for model development and a validation/test subset for internal performance evaluation. No case was used in more than one subset.

3) Terminology clarification:

To improve clarity, we have renamed the cohorts in the revised manuscript (Supplementary Figure 1, please refer to Lines 486-497 in the revised manuscript): The former “internal (training subset)” is now referred to as “Training cohort”. The former “internal (testing subset)” is now referred to as “Internal test cohort”.

4) Patient overlap statement:

We have explicitly added a statement in the revised Methods section noting that no patient overlap occurred across any cohorts or subsets (please refer to Lines 119-123 in the revised manuscript).



Revised Supplementary Figure 1 Brief usage summary of the internal set and the two external sets

A deep learning model was first developed and internally validated using 512 slides from a single medical center (Training Cohort), achieving an AUROC of 0.989. The developed G-NECNet was then validated externally on the testing subset of the internal cohort of 665 slides, on the external cohort of 344 slides and on the consultation cohort of 113 slides, yielding an AUROC of 0.993, 0.985 and 1.000, respectively. Internal-cohort, the internal dataset from the Sun Yat-sen University Cancer Center; External-cohort, external dataset from the Sixth Affiliated Hospital of Sun Yat-sen University; Consultation-cohort, external dataset from multiple medical centers; AUROC, area under the receiver operating curve. No patient overlap occurred across any cohorts or subsets.

Within the Method section:

“These datasets included the Internal-cohort sourced from the Sun Yat-sen University Cancer Center (SYSUCC), the External-cohort sourced from the Sixth Affiliated Hospital of Sun Yat-sen University (SYSU-6th Hospital), and the Consultation-cohort from an external consultation database. Although SYSUCC and SYSU-6th Hospital are affiliated with Sun Yat-sen University, they are independent institutions with separate patient registries and no patient overlap occurred across any cohorts or subsets.”

3. Given the rarity of gastric NET, further description is needed regarding how cases in the external cohort and consultation cohort were selected. If neuroendocrine tumors, other rare histological types, or non-neoplastic diseases were excluded, this should be clearly stated.

Our reply:

We appreciate your insightful comment. To clarify, we described in detail the exclusion criteria for the three cohort in the method section. These exclusion criteria were applied consistently across both the internal, external and consultation datasets to eliminate potential confounders and ensure that model training and validation specifically targeted the differentiation between G-NEC and GC. (please refer to Lines 128-134 in the revised manuscript)

Within the Method section:

“All histopathological diagnoses were independently reviewed and confirmed by two board-certified gastrointestinal pathologists according to the 2022 WHO Classification of Tumors of the Digestive System (WHO 5th Edition)^[25]. To ensure diagnostic purity, we excluded the following categories: (1) gastric neuroendocrine tumors (G-NET, grades 1-2); (2) mixed neuroendocrine-non-neuroendocrine neoplasms (MiNEN); (3) other rare histologic subtypes such as lymphoma, gastrointestinal stromal tumor (GIST), and metastatic tumors; (4) non-neoplastic gastric lesions (e.g., chronic gastritis, ulcer, or hyperplasia).”

4. The GitHub link provided in line 292 is not functioning, making data availability currently unknown.

Our reply:

We extend our sincere gratitude for your invaluable insights. To address your concerns, we have updated the link, uploaded the source data as an xlsx file and completed the data accessibility statement (please refer to Lines 321-330 in the revised manuscript and Source Data).

Minor Comments

1. Please provide the definition of NEC diagnosis used in this study (i. WHO 5e).

Our reply:

Thank you for the excellent suggestion. The definition of NEC diagnosis has been added in the method section. (please refer to Lines 145-152 in the revised manuscript)

Within the Method section:

“According to the WHO Classification of Tumours of the Digestive System (5th Edition, 2022), the diagnosis of G-NEC requires histological evidence of poorly differentiated carcinoma, typically demonstrating solid, trabecular, or sheet-like growth patterns, extensive necrosis, and marked nuclear atypia with a high nuclear-to-cytoplasmic ratio. Neuroendocrine differentiation must be confirmed by immunohistochemical staining positive for at least one neuroendocrine marker, such as synaptophysin (Syn), chromogranin A (CgA), or insulinoma-associated protein 1 (INSM1). A high proliferative rate (defined as a mitotic count exceeding 20 mitoses per 10 high-power fields and/or a Ki-67 proliferation index greater than 20%) is also required for diagnosis.”

2. Please clarify whether the pathologists’ diagnoses in the reader study were made using only H&E stains.

Our reply

Thank you for the excellent suggestion. The clarification has been added in the method section. (please refer to Lines 206-208 in the revised manuscript)

Within the Method section:

“Blinded to the diagnosis and clinical information, pathologists reviewed only H&E-stained histopathological slides and classified each case into G-NEC or gastric adenocarcinoma based on the histopathological features of G-NEC and their expertise.”

3. Lines 162-163 mention the number of tiles used. Does this refer to the number of whole slide images used? Also, please verify that the numbers allocated for training and validation are not reversed.

Our reply

Thank you so much for your insightful and detailed comments. Lines 162-163 mention the number of tiles is the number of whole slide images. We have carefully revised the corresponding descriptions in the manuscript and clarified description to eliminate potential confusion. The detailed clarifications are as follows:

“About 16 G-NEC slides and 85 GC slides were obtained for the training of each classifier, and about 61 G-NEC slides and 340 GC slides for internal validation. For the training of each classifier, the slides of tumor regions selected by the tumor detector and their corresponding labels (1 for "G-NEC" and 0 for "GC") were used respectively as the inputs and the outputs of G-NEC”. *(please refer to Lines 186-190 in the revised manuscript)*.

Reviewer #2 (Remarks to the Author):

This manuscript aims to develop a deep learning-based diagnostic that serves as a rapid and cost-effective solution to aid in the diagnosis of gastric neuroendocrine carcinoma (G-NEC). The study was well-designed and logically structured, and the text was concisely written.

Strengths:

1. Three cohorts were included in the study: internal, external, and consultation-cohorts. The slides and images were collected from multiple medical centers.
2. The 's performance was compared with that of pathologists with different levels of expertise/experience, and it shows that the 's diagnosis surpasses the pathologists.
3. The inclusion of workflow figures, detailed statistics of the study participants, performance during five-fold validation, and histology images is very friendly to readers.
4. The study results support the hypothesis that the deep learning-based diagnostic overcomes the routine diagnosis of G-NECs with the assistance of IHC and molecular tests.

Our reply:

We were pleased that the reviewer felt that “the study was well-designed and logically structured” and “the text was concisely written”.

Some specific points:

Improvements:

1. The main diagnostic challenge is to differentiate G-NEC from poorly differentiated gastric adenocarcinoma. The differentiation grades were listed in Supplementary Table 1, including poorly, moderately, and well-differentiated gastric adenocarcinoma. For the unsuccessful cases predicted by the G-NECNet, did the author review the histology to see if the cases had more ambiguities/overlapping features with gastric adenocarcinoma?

Our reply:

We appreciate the editor’s insightful comment regarding the diagnostic ambiguity between G-NEC and poorly differentiated gastric adenocarcinoma. To address this point, we re-examined the histology of cases misclassified by G-NECNet and report the results in Supplementary Table 3(please refer to Lines 525-527 in the revised

manuscript).

As shown in Supplementary Table 3, the proportion of poorly differentiated tumors was slightly higher among the misclassified cases than among the correctly diagnosed ones across all cohorts (e.g., 75.0% vs. 57.5% in the internal cohort; 85.7% vs. 63.8% in the external cohort). Although the differences did not reach statistical significance by Fisher’s exact test ($P = 0.256$ and 0.056 , respectively), these data suggest a tendency that poorly differentiated tumors are more prone to misclassification.

Supplementary Table 3 Association between tumor differentiation and G-NECNet diagnostic performance across different cohorts

G-NECNet Diagnosis status	Successful		misclassified		<i>P</i>
	Poor differentiation	Middle and well differentiation	Poor differentiation	Middle and well differentiation	
Internal test cohort	468 (57.49%)	346 (42.51%)	9 (75.00%)	3 (25.00%)	0.256
External-cohort	203 (63.84%)	115 (36.16%)	18 (85.71%)	3 (14.29%)	0.056
Consultation-cohort	71 (63.39%)	41 (36.60%)	1 (100.0%)	0 (0.00%)	1.000

Note: The number of participants with correct diagnoses and those with incorrect diagnoses was compared by Fisher exact test. The *P* value was evaluated from a two-sided test.

2. Did the study include any cases with combined components (both neuroendocrine carcinoma and adenocarcinoma), and how about the performance for such cases?

Our reply:

We appreciate your insightful comment. To avoid confounding the binary classification task, we excluded mixed neuroendocrine-non-neuroendocrine neoplasms (MiNEN) and cases with clearly combined NEC and adenocarcinoma components during case selection (please refer to Lines 131-137 in the revised manuscript). Therefore, the present study did not evaluate model performance on mixed-component cases. The model’s performance on such mixed lesions remains unknown and should be examined in future work, as they represent an important diagnostic subgroup. As suggested, a paragraph discussing this limitation has been added (please refer to Lines 302-308 in the revised manuscript).

Within the Method section:

“All histopathological diagnoses were independently reviewed and confirmed by two board-certified gastrointestinal pathologists according to the 2022 WHO Classification of Tumors of the Digestive System (WHO 5th Edition)^[25]. To ensure diagnostic purity, we excluded the following categories: (1) gastric neuroendocrine tumors (G-NET, grades 1-2); (2) mixed neuroendocrine-non-neuroendocrine neoplasms (MiNEN); (3) other rare histologic subtypes such as lymphoma, gastrointestinal stromal tumor (GIST), and metastatic tumors; (4) non-neoplastic gastric lesions (e.g., chronic gastritis, ulcer, or hyperplasia).”

Within the Discussion section:

“First, although this was a multicenter study, the overall sample size remained relatively limited, and the dataset included only histologically confirmed G-NEC and gastric GC cases. This design was chosen because differentiating G-NEC from poorly differentiated GC represents the most challenging and clinically relevant diagnostic scenario in gastric pathology. However, this binary classification setting does not encompass the full spectrum of gastric lesions, such as MiNEN, other rare histological types, and non-neoplastic conditions. Expanding future studies to include a broader population would enable true “detection” of G-NEC and enhance model generalizability.”

3. Please check the figure 3 legend, and panel d was not included.

Our reply:

We appreciate your careful review and have revised the description in the figure 3 legend. (please refer to Lines480 in the revised manuscript).